

The ESP32 will then send the query to ChatGPT, and the output obtained from ChatGPT will be reproduced on the same TFT screen. You don't need an elaborate computer for this task. Of course, you need an internet Wi-Fi connection

TABLE 1
BILL OF MATERIALS

Items	Quantity
ESP32 MCU (MOD1)	1
PS2 keyboard	1
8.9cm TFT (MOD2)	1
Wires, PCB, connector	
7805: 5-volt regulator IC (IC1)	1
Diode 1N4007	1

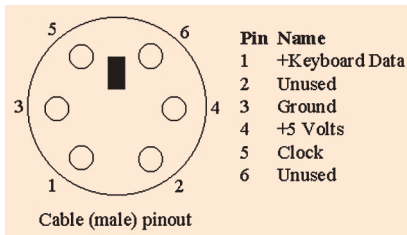


Fig. 3: PS/2 key and mouse cable details

and a secret key to access the ChatGPT API. This is the single-point access key for openai.com; no other login or password is required.

The author's prototype is shown

TABLE 2
CONNECTION OF TFT AND ESP32

TFT Pin	ESP32 Pin
CS	33
DC	15
RST	32
WR	4
RD	2
D0	12
D1	13
D2	26
D3	25
D4	16 (RX2)
D5	17 (TX2)
D6	27
D7	14
VCC/5V	5 volts from regulator
GND	Ground

in Fig. 1. The components used in the terminal are listed under Bill of Materials in Table 1.

Circuit and working

Fig. 2 shows the circuit diagram of the ESP32-based ChatGPT terminal. It is built around the ESP32 board (MOD1), an 8.9cm (3.5-inch) TFT display (MOD 2), 5V voltage regulator IC 7805 (IC1), PS2 keyboard, and a few other components.

For the proper connection of the entire ChatGPT terminal, connect the TFT display to the ESP32 board as well as the PS2 keyboard pins on the

TABLE 3
CONNECTION OF PS2 KEYBOARD PIN AND ESP32

PS2 Keyboard Pin	ESP32 Pin
Data pin	35
IRQ pin (clock pin)	34
5V	5 volts from regulator
GND	Ground

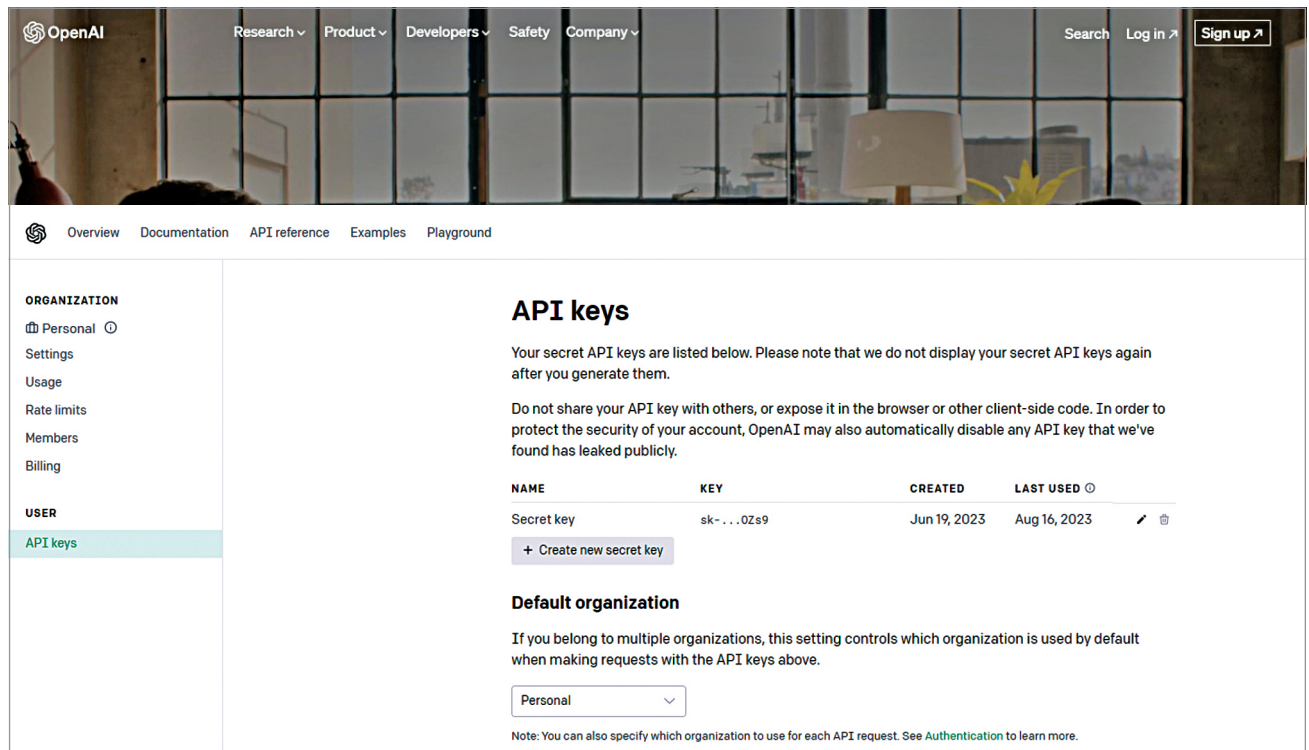


Fig. 4: OpenAI API creation